When Does Privileged Process Supervision Help? A Teacher-Level Sweet Spot, a Verified Transfer Null, and a Supervision-Mismatch Diagnosis

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Abstract

Process reward models (PRMs) score intermediate reasoning steps, and many pipelines construct their step-level supervision automatically from a stronger teacher model. We study two questions: when does privileged (answer-aware) supervision help such a teacher, and separately, can that advantage be distilled into a small, deployable, answer-blind student verifier? On a controlled three-tier difficulty probe (GSM8K, MATH, OlympiadBench), a privileged teacher's step-error localization improves only in a tractability sweet spot: privilege is redundant on easy problems the teacher can already self-verify (GSM8K, n.s.), unusable on the hardest problems where the reference is too long to exploit (OlympiadBench, n.s.), and helpful precisely in between (MATH, +0.051 F1, 95% CI $[0.01, 0.09]$), replicating across model families (Qwen-27B, +0.082 F1) and within MATH by difficulty level. This advantage does not distill: a 1.5B student trained on the privileged teacher's labels is statistically indistinguishable from one trained on no-GT labels (ΔROC-AUC -0.010, 95% CI $[-0.025, 0.005]$), and neither beats majority vote on downstream best-of-N re-ranking. We show this null is not explained by pipeline insensitivity—a positive control confirms the training path tracks teacher quality (+0.050 ROC-AUC, 95% CI $[0.021, 0.078]$, significant)—nor primarily by student capacity: under an identical training and evaluation path, a 3B verifier trained on format-matched ProcessBench-style gold labels reaches 0.73–0.79 ROC-AUC across eight seeds on full ProcessBench-MATH (up to 0.83 with post-hoc score averaging), while the same generated teacher labels remain weak (0.55–0.63 ROC-AUC) at both 1.5B and 3B scale. The evidence points to a supervision distribution/format mismatch, not raw capacity, as the dominant failure mode, and offers a practical diagnostic: test whether format-matched supervision can support the target verifier before scaling students or searching over training recipes.

1 Introduction

Process reward models assign scores to intermediate reasoning steps and are increasingly used to supervise or verify mathematical reasoning systems. Because human step-level annotation is expensive, many PRM pipelines construct supervision automatically, either by giving a labeling teacher privileged (answer-aware) access to a reference solution, or by other weak-supervision means. Two questions follow naturally, and most prior work answers only the first: (1) does privileged access actually make the teacher a better labeler, and if so, when; and (2) does that advantage survive distillation into a small, deployable, answer-blind student verifier?

We answer both under a controlled setup that holds the student architecture, optimizer, and evaluation benchmark fixed while varying exactly one thing at a time. First, we characterize *when* privileged supervision helps a teacher (§4): it is not uniformly useful, but pays off only in a tractability window where the teacher both needs the reference (it cannot already self-verify)

and can act on it (the reference is short enough to follow). Second, we ask whether this real, mechanistically-understood teacher-level advantage distills into a small student PRM (§5): it does not, and we show this null is not an artifact of an insensitive training pipeline. Third, having ruled out pipeline insensitivity, we ask the natural follow-up—is the failure to distill a student-capacity problem or a supervision problem? (§6). Under the identical training and evaluation path, we show that format-matched gold process supervision *does* produce a competent verifier, while our generated teacher labels do not, at both 1.5B and 3B student scale. This decomposes what would otherwise be a flat, unsatisfying null into three connected findings: a real, well-characterized teacher effect; a verified failure of that effect to distill, with pipeline sensitivity confirmed; and a concrete diagnosis of why it fails, pointing at supervision distribution and format rather than capacity.

2 Related Work

Process supervision and process reward models. A line of work has established that supervising *how* a model reasons, step by step, is more effective than supervising only whether the final answer is correct. Lightman et al. [1] compared the two regimes directly and found that process supervision substantially outperforms outcome supervision on the challenging MATH dataset, releasing the large PRM800K corpus of step-level human labels. Because human step annotation is expensive, Math-Shepherd [2] showed that process supervision can be constructed automatically—assigning a reward to each solution step without manual labels—and used the resulting signal both to re-rank candidate solutions and to drive step-level reinforcement learning. These PRMs treat each step's correctness as a scalar and deploy the PRM as a frozen scorer or ranker. We share the step-level granularity but depart from the scalar view for the teacher-level analysis: we treat the teacher's step judgment, including its natural-language critique, as the object to be distilled rather than a number to be predicted.

Generative PRMs. A more recent thread makes the verifier itself generative, producing a reasoning trace before it judges. GenPRM [3] performs explicit chain-of-thought reasoning with code verification before scoring each step, while ThinkPRM [4] fine-tunes a long-CoT verifier that articulates a rationale for every step and reaches strong ProcessBench performance from only a few thousand process labels. These models are close to ours in spirit—they generate critiques rather than emit bare scores—but they are designed as verifiers evaluated at inference time, not as a teacher whose score-plus-critique behavior is distilled into a separate, smaller, answer-blind student. None isolates *when* privileged judgment actually helps, which is the question we take up in §4.

Weak and generated process supervision. Recent weak-supervision approaches train PRMs without dense human process labels, using outcome labels, generated rationales, or model-derived signals [10]. These methods make PRM training more scalable, but they raise the question we study in §6: when generated labels fail, is the problem the student, the recipe, or the supervision distribution? ProcessBench itself frames earliest-error identification as a benchmark on which existing PRMs often fail to generalize across sources [7], and Qwen2.5-Math PRMs are strong public baselines trained at much larger scale that provide an external calibration point for our results [8, 9]. Concurrent work studies PRM transfer across GSM8K, MATH, OlympiadBench, and OmniMath more directly—ProcessLID reports explicit transfer tables across these sources [11], and RetrievalPRM frames PRM failures as question/step out-of-distribution issues across the same sources [12]. Our contribution relative to this thread is not that cross-source transfer is novel, but

the controlled contrast between generated teacher labels and ProcessBench-style gold labels under one fixed student, optimizer, and evaluation path (§6).

Expert–amateur contrast. Our framing descends from work that extracts signal from the gap between a stronger and a weaker model. CLEAR [5] contrasts feedback from a larger "expert" and a smaller "amateur" model and applies the refined difference to iteratively improve outputs across reasoning tasks; it is the direct predecessor of the present project. LightReasoner [6] similarly exploits the behavioral divergence between a strong expert and a weak amateur to pinpoint high-value reasoning moments and distill them, notably without ground-truth labels. We retain the ground-truth-free constraint at student test time but invert the axis of interest: rather than contrasting models of different capacity, we study a teacher with a privileged *view*—access to the reference solution while labeling— distilled into a student that is answer-blind by construction. The relevant axis for us is information, not size.

Data-centric learning and privileged information. Data-centric AI argues that systematic changes in data quality can dominate model-side changes [13]. Our §6 result is a PRM-specific instance: a fixed small verifier's performance changes substantially with the supervision source, holding the model and optimizer fixed. The privileged-label experiments in §4–§5 are also related to learning using privileged information [14], but with an important boundary: privileged information can improve the teacher's per-example inference without producing labels that distill into a better answer-blind student.

Our position. We make three contributions, and we are deliberate about which is load-bearing. **First** (§4), we characterize *when* privileged supervision is worth anything at the teacher level: a privilege × difficulty × richness effect in which a teacher's answer-aware judgment improves step-error localization only when the problem is hard enough that the teacher cannot self-verify *and* the privilege is rich enough to act on. To our knowledge this conditional structure has not been isolated in the PRM or expert–amateur literature; it is the validated result and the paper's spine. **Second** (§5), we show this advantage does not distill into a small student verifier, and we rule out the most obvious confound—that the training pipeline simply cannot detect any teacher-quality difference—with a positive control. **Third** (§6), we diagnose the failure further: under an identical training/evaluation path, format-matched gold process supervision produces a competent verifier while our generated labels do not, which points at supervision distribution and format rather than student capacity as the dominant explanation.

3 Experimental Setup

Teacher-level privilege probe (§4). We evaluate a teacher's ability to localize the first erroneous step in a multi-step solution under three conditions that differ only in the privileged information available at labeling time: (i) **no-GT**, an answer-blind condition in which the teacher sees only the problem and the candidate solution; (ii) **+answer**, adding the final reference answer; and (iii) **+solution**, adding the full worked reference solution. Performance is reported as ProcessBench-style first-error F1, together with error-detection accuracy. Reported deltas are taken against the no-GT condition.

Student distillation (§5). The student is `Qwen2.5-1.5B-Instruct` with a lightweight score-and-critique head, trained via LoRA on teacher-produced score and natural-language critique labels

Teacher	F_1	correct acc	error acc	parse fail
Gemma-4-26B (primary)	**0.909**	1.000	0.833	0.000
Qwen-27B	0.873	0.917	0.833	0.000
Qwen-35B-A3B	0.839	1.000	0.722	0.019
Qwen-4B-Thinking	0.000	1.000	0.000	0.846

Table 1: Teacher bake-off, with-GT ceiling (N=30).

(N_{train}=1000, N_{eval}=400, 2 epochs). We compare students trained on privileged (+solution) versus no-GT teacher labels, plus a privileged-score-only ablation that drops the critique loss.

Distribution-matched diagnostic (§6). To separate student capacity from supervision distribution/format, we run a second, matched training/evaluation path on `Qwen2.5-3B-Instruct` with a score-only head, LoRA adapters, binary cross-entropy loss, balanced batches, and 500 training steps. All headline results in §6 evaluate on the same shuffled ProcessBench-MATH split of 1,000 solutions (6,505 steps, 594 error steps) with exact serial scoring. We compare two supervision sources under this identical path: (a) ProcessBench-style gold step labels from the first 400 examples of a non-MATH source configuration (GSM8K or OmniMath), flattened to per-step labels, and (b) our generated teacher labels from the privileged/no-GT labeling pipeline above. Exact problem-string overlap between each training source and the MATH-1000 evaluation split is zero.

4 The Privilege Sweet Spot (Teacher Level)

4.1 Teacher selection

Before probing the effect of privilege, we fix the teacher by an upper-bound bake-off in which every candidate is granted ground-truth access (the with-GT ceiling, N=30). Table 1 shows the official Gemma-class checkpoint attains the strongest first-error F1 (0.909) with perfect format compliance and the highest throughput; a comparably-sized Qwen variant follows closely (0.873). A reasoning-tuned candidate collapses under the structured-output protocol, failing to parse on the large majority of instances—a reminder that verbal reasoning fluency does not by itself confer reliable structured step adjudication. We adopt the Gemma checkpoint as the primary teacher and retain the Qwen variant for the cross-family replication.

4.2 The sweet spot

Our central finding is that privileged supervision is *conditionally* beneficial: it helps only when the problem is hard enough that the teacher cannot self-verify and tractable enough that the teacher can still exploit the reference. The solution-privilege gap is non-significant on easy problems (GSM8K, -0.052, n.s.), opens to a significant $+0.051$ F1 on hard-but-tractable problems (MATH, 95% CI $[0.01, 0.09]$), and returns to non-significant on the hardest tier (OlympiadBench, -0.03, n.s.). The three points trace an inverted U in difficulty (Fig. 1a). We do not lean on the two non-significant endpoints alone to assert a curve: the same inverted U reappears *within* MATH by difficulty level (Fig. 1b), peaking at the intermediate level 3 ($+0.11$, n=91) and collapsing at the easiest level 1 (-0.13, n=32), and it is corroborated by the mechanism analysis of §4.4, where the rescue rate declines monotonically with reference length. The reading the data support is conjunctive: privilege pays off only where the teacher both needs the reference and can use it.

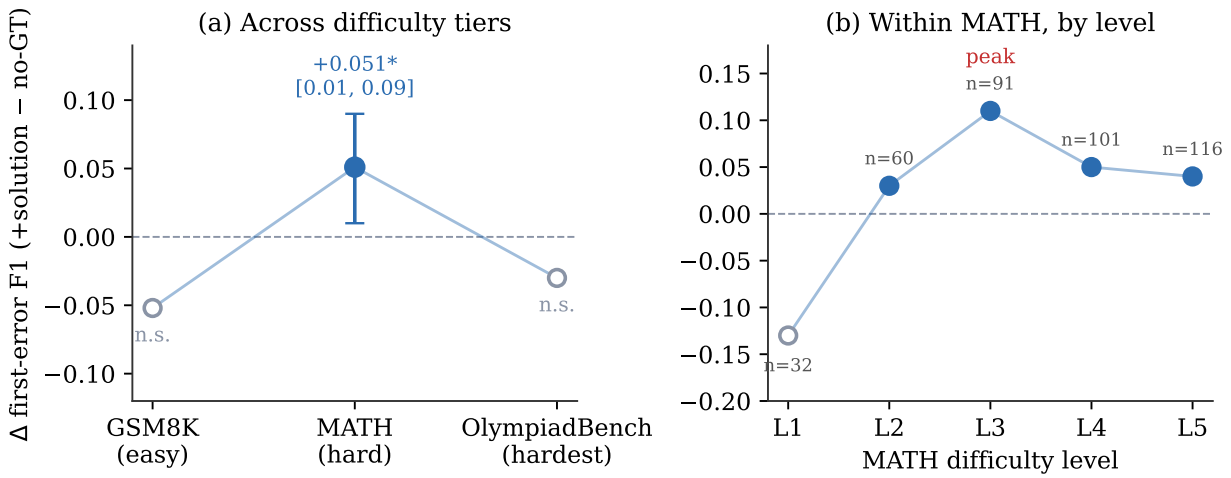


Figure 1: The privilege sweet spot is an inverted U in difficulty. **(a)** Solution-privilege gap (Δ first-error F1, +solution minus no-GT) across the three difficulty tiers (Gemma teacher): non-significant on the easy and hardest tiers, significant only on the hard-but-tractable middle tier (MATH, +0.051, 95% CI $[0.01, 0.09]$). **(b)** The same curve reappears within MATH by difficulty level (N=400), peaking at level 3 and collapsing at level 1—the effect is not an artifact of comparing three heterogeneous datasets.

Tier	Benchmark (teacher)	no-GT	+answer	+solution	Δ (solution)
Easy	GSM8K (Gemma, N=50)	0.889	0.856	0.837	−0.052 (n.s.)
Hard	MATH (Gemma, N=400)	0.726	0.726	0.777	+**0.051** $[0.01, 0.09]$
Hardest	OlympiadBench (Gemma, OE)	–	–	–	−0.03 (n.s.)
Hard	MATH (Qwen-27B, N=150)	0.653	0.690	0.735	+0.082

Table 2: The privilege sweet spot across difficulty tiers, with cross-family replication. Values are first-error F1. MATH (Gemma) is the N=400 definitive run (95% CI shown); OlympiadBench is OE-only, verified.

Table 2 reports the three tiers in full, together with the Qwen-27B cross-family replication. On GSM8K, neither the bare answer nor the full solution improves an already-saturated judgment; the teacher is, in effect, its own oracle. On MATH, where unaided self-verification degrades, the full worked solution lifts first-error F1 from 0.726 to 0.777 (+0.051, 95% CI $[0.01, 0.09]$) and error recall from 0.608 to 0.661. OlympiadBench, harder still, returns the gap to ≈ 0: the references are too long for the teacher to follow, so the privilege it is handed is no longer usable. Critically, the cross-family Qwen-27B teacher reproduces the effect on MATH (+0.082 F1, error detection $0.526 \rightarrow 0.667$), substantially weakening any single-model-artifact explanation.

4.3 Richness, not mere access

The MATH panel isolates a second, sharper effect. The +answer condition—a correct final answer but no reasoning—is indistinguishable from the answer-blind baseline: the two rows are identical, and the bare answer flips zero predictions. Only the full worked solution moves the metric. The operative variable is therefore not whether the teacher is privileged, but whether the privilege is *rich*: a reference reasoning trace supplies the step-by-step scaffold needed to localize an error,

Problem (MATH level)	What no-GT missed	+solution feedback at the flagged step
Tens digit of $7! + 8! + \cdots + 2006!$ (L3)	Predicted no error anywhere (missed step 3)	"Incorrectly assumes the tens digit of a sum is the sum of the tens digits; fails to account for a carry from the units place."
Regular-tetrahedron centroid ratio (L5)	Flagged step 1 instead of the actual error at step 4	"Incorrectly concludes $PQ/AQ = 1/3$ from a 3:1 median claim; in a tetrahedron the centroid gives $AP/PQ = 3{:}1$, not PQ/AQ."
Base-6 representation of 355 (L2)	Predicted no error anywhere (missed step 0)	"The division $355 \div 6$ is incorrect; $355 = 59 \times 6 + 1$, so the remainder should be 1."

Table 3: Three rescue cases (gold error step known, no-GT prediction wrong, +solution prediction correct), selected to span error types: a missed carry/place-value step, a misapplied geometric ratio, and a missed arithmetic slip. 37 such rescue cases exist in the N=400 evidence pack; these three are illustrative, not chosen for effect size.

whereas a final answer, which the teacher cannot back-propagate into the faulty step, supplies nothing it can act on.

4.4 Mechanism: privilege rescues self-verification failures

The sweet spot is not a raw difficulty effect; it is a rescue of self-verification failures, decomposing into two gates (N=400). **Gate 1 (the teacher must need help).** Of 227 problems containing an error, the no-GT teacher misses 89. Adding the worked solution rescues 29 of those (33%), while breaking only 17 of the 138 it had already caught (12%)—the benefit concentrates exactly where self-verification fails. **Gate 2 (the teacher must be able to use it).** The rescue rate falls monotonically with the length of the reference solution (short 0.37 → mid 0.33 → long 0.28), which is the mechanism behind the collapse at the hardest tier. Together the gates identify the moderator as self-verification-failure × tractability, precisely the inverted U of Fig. 1. We note one caveat we cannot fully resolve here: at the hardest tier, "too long to use" conflates mathematical intractability with long-context attention degradation over very long references; both are consistent with an unusable reference, and separating them is future work.

5 Does the Teacher Advantage Transfer to the Student?

The teacher-level sweet spot in §4 is the validated result. The natural next question is whether it distills: if we train a small (1.5B) student PRM on the privileged teacher's score-plus-critique labels, does that student inherit a usable advantage over one trained on no-GT labels? Across a validated N=1000 run (Gemma-4 labeling confirmed through ∼32k served requests), the answer is no. We report this as a clean, diagnosed negative.

5.1 Step-level and downstream transfer

We compare three students—privileged+critique, privileged score-only, and no-GT+critique—on threshold-free ROC-AUC, the appropriate measure here because first-error F1 at a fixed score-head cutoff moves with calibration rather than capability. Table 4 (top) shows the no-GT student

Step-level (N_{train}=1000, N_{eval}=400)				
Student (teacher labels)	ROC-AUC	PR-AUC	F_1	error recall
Privileged + critique	0.631	0.130	0.170	0.357
Privileged, score-only	0.624	0.135	0.184	0.427
No-GT + critique	**0.641**	0.121	0.198	0.467
Downstream best-of-N (N=1000 problems, 8 candidates)				
Verifier	pass@1	prm rerank	majority vote	oracle pass@N
No-GT student	0.337	**0.373**	0.391	0.517
Privileged student	0.335	0.349	0.382	0.518

Table 4: Student transfer. Privilege does not transfer: the no-GT student is equal-or-better at step level (paired-bootstrap $\Delta = -0.010$, 95% CI $[-0.025, 0.005]$) and re-ranks better downstream, and neither verifier beats majority vote.

is highest on ROC-AUC (0.641 vs. 0.631), and the run is non-degenerate (predicted-error rate ≈ 0.3 across all cells). A paired bootstrap over steps (10,000 resamples) puts the privileged−no-GT gap at -0.010 ROC-AUC, 95% CI $[-0.025, 0.005]$, two-sided p=0.20: the two students are statistically indistinguishable, with the point estimate favoring no-GT.[1]

The motivating promise of a ground-truth-free localizer is a test-time gain: given N sampled solutions, the student scores each and we re-rank. Table 4 (bottom) reports the downstream best-of-N result on N=1000 problems (8 candidates, `gemma-2-9b-it` generator, an *independent* candidate pool per verifier). The no-GT verifier re-ranks *better* than the privileged one (0.373 vs. 0.349), and neither beats the majority-vote baseline (≈ 0.39); the oracle (0.517) shows the headroom is real, so the failure is the verifiers', not the candidate pool's.

5.2 Why the null: a diffuse, not absent, teacher gap

The negative is not that privileged and no-GT labels are identical. A probe of the actual Gemma-4 labeling teacher reproduces a real +0.07 solution gap (F1 0.716 → 0.786; error recall 0.577 → 0.654, N=150 MATH), consistent with §4. The issue is the *shape* of the gap. Over 6,340 labeled steps, privileged and no-GT labels agree on only 69%; on the $\sim$31% where they differ, the disagreement is nearly symmetric—privilege flags an error where no-GT does not on 1,001 steps, while no-GT flags one where privilege does not on 956. The teacher's privilege churns labels in both directions; its small net quality gain is diffuse, not a clean directional signal a 1.5B student can latch onto. A same-pool paired best-of-N is consistent with this: at N=200 the privileged and no-GT re-rankers are indistinguishable (McNemar p=0.14; 5 vs. 12 discordant pairs), both below majority vote. That paired test uses a *shared* candidate pool and is underpowered (17 discordant pairs); a properly powered N=1000 shared-pool paired test is left to future work, and we do not draw the null from it—the load-bearing null is the independent-pool result of Table 4.

5.3 Positive control: the pipeline is sensitive to teacher quality

The results above leave one obvious confound open: does the training pipeline fail to distill privilege specifically, or is it simply insensitive to *any* teacher-quality difference? We test this directly. We relabel the same candidate solutions with a deliberately weaker labeling model

[1]This is a paired bootstrap over steps (steps within a solution are correlated, so the interval is mildly anti-conservative); both the sign and the zero-crossing interval support the null.

(`Qwen2.5-3B-Instruct`, in place of the 26B Gemma-4 teacher) and train an identically-configured 1.5B student. This weak-teacher student reaches ROC-AUC 0.575 on the full 1,000-problem ProcessBench-MATH split, against 0.625 for the Gemma-4-labeled privileged+critique student on the identical 6,505-step split.[2] A paired bootstrap (10,000 resamples) on this gap gives a mean difference of +0.050 ROC-AUC, 95% CI $[0.021, 0.078]$, two-sided p=0.001—significant and excluding zero. (The weak-teacher student's score head is also markedly less calibrated, predicting "error" on only 1.6% of steps, so part of this gap reflects degeneracy rather than a smooth quality gradient; we report it as-is.)

This is the key validity result for §5: because student quality tracks teacher quality when the teacher-quality difference is large and unprivileged (weak vs. strong labeling model), the pipeline is demonstrably capable of transmitting a teacher-quality signal into the student. The null in §5.2 is therefore not explained by pipeline insensitivity; it is specifically the privileged-vs-no-GT distinction—a subtler, more diffuse signal—that fails to distill at this student scale. This motivates the next question directly: is the remaining gap a matter of student capacity, or of the generated-label supervision itself?

6 Diagnosing the Null: Capacity or Supervision?

Having ruled out pipeline insensitivity (§5.3), is the failure of privilege to distill because the 1.5B student lacks the capacity to represent the signal, or because the generated-label supervision itself is poorly matched to what ProcessBench-MATH needs? We address this with a controlled ablation on `Qwen2.5-3B-Instruct` (§3): holding the verifier architecture, training path, and evaluation benchmark fixed, we vary only the source and format of step-level supervision.

6.1 Format-matched gold labels transfer; generated labels remain weak

Table 5 shows the main result. With the student, training path, and evaluation held fixed, ProcessBench-style labels from GSM8K and OmniMath transfer strongly to full ProcessBench-MATH (0.75/0.77 ROC-AUC), while generated teacher labels remain weak (0.55/0.63)—even when the generated-label baseline is chosen as the strongest checkpoint from a broader preliminary sweep (appendix). Every GSM8K and OmniMath seed significantly beats both generated-label baselines under a sequence-cluster bootstrap (resampling whole solutions, whose steps are correlated); against the best generated-label baseline, gaps range from +0.11 to +0.12 ROC-AUC across the strongest seeds and up to +0.20 against the privileged-BCE baseline, all two-sided p=0.001 (full per-seed table in the appendix). Held-out threshold calibration (threshold chosen on the first 200 MATH sequences, evaluated on the remaining 800) agrees with the ranking: gold seeds reach calibrated F_1 of 0.31–0.33 against 0.17/0.21 for the generated-label baselines.

6.2 The source distribution still matters

This is not a blanket claim that any non-MATH process labels transfer cleanly to MATH. OlympiadBench is high-variance in our runs: two seeds give 0.585 and 0.716 ROC-AUC on full MATH-1000 (mean 0.651), one significantly below and one significantly above the best generated-label baseline. We treat OlympiadBench as a boundary diagnostic rather than a third clean headline source; the pattern still supports a source-distribution view, but the split matters.

[2]The 0.625 here is evaluated on the full 1,000-problem split, versus the 400-problem snapshot behind Table 4's 0.631; we recompute the control directly from the committed per-step scores of both conditions so it is independently reproducible.

Training source	Seeds	ROC-AUC	PR-AUC	Best F_1^*	Calib. F_1
GSM8K gold → MATH1000	0–3	0.7515 (0.7256–0.7760)	0.2188	0.3144	0.3166
OmniMath gold → MATH1000	0–3	0.7694 (0.7539–0.7869)	0.2524	0.3276	0.3062
Qwen2.5-Math-7B-PRM800K (public)	–	0.8379	0.3254	0.3991	0.3855
Generated privileged labels, BCE	0	0.5503	0.0992	0.1803	0.1673
Generated no-GT labels, rank loss	0	0.6324	0.1418	0.2221	0.2085

Table 5: Full ProcessBench-MATH (N=1000) transfer under an identical Qwen2.5-3B score-head training/eval path. Parentheses show seed ranges. *Best F_1 is threshold-swept on the eval slice (diagnostic); Calib. F_1 is held-out (threshold on the first 200 sequences, evaluated on the remaining 800). Qwen2.5-Math-7B-PRM800K is an external calibration baseline, not a matched recipe—this is not a SOTA PRM claim.

Diagnostic	Condition	ROC-AUC
Score head, matched MATH 200/200	Gold labels	0.7129
Frozen probe, generated teacher labels	Privileged	0.5497
Frozen probe, generated teacher labels	No-GT	0.5559
Frozen probe, ProcessBench-gold 200/200	Gold labels	0.6308

Table 6: Diagnostics separating score-head/representation capacity from supervision mismatch.

6.3 The failure is not the score-head machinery

A plausible concern is that the 3B score-head is simply too weak to learn step-level verification. Table 6 argues against this. Training the same path on the first 200 shuffled MATH examples and evaluating on the next 200 reaches 0.7129 ROC-AUC—not a benchmark claim (train and eval share a family), but evidence that the machinery can learn the target when labels and distribution are aligned. Frozen-representation probes tell the same story: class-balanced logistic probes on frozen Qwen2.5-3B step-boundary embeddings trained on generated teacher labels transfer weakly (0.52–0.56), while the matched ProcessBench-gold probe reaches 0.631. The supervision distribution is doing meaningful work.

6.4 Training-recipe changes did not rescue generated labels

Before reframing the result as a supervision mismatch, we tested a range of score-head and distillation variants on the generated labels (appendix). Naive score-plus-critique distillation was unstable until a feedback-token truncation bug was fixed; after the fix, equal-weight critique distillation remained weak or negative for privileged labels. Pure ranking loss and BCE-plus-rank did not rescue privileged transfer—they often favored the no-GT labels instead—and longer BCE training increased variance without producing a competent MATH verifier. These negative ablations do not prove no generated-label method can work, but combined with the 1.5B ablations of §5 (soft-BCE, hard-verdict, and logit-KD distillation, same result), they reduce the likelihood that the gap is a single bad optimizer setting at a single student scale.

6.5 Interpretation

Taken together: the training pipeline is sensitive to teacher quality (§5.3); the score-head machinery and student representation can learn a competent verifier when supervision is distribution-matched (§6.1–6.3); and a broad sweep of loss functions and student scales fails to rescue the generated-

label signal specifically. The most parsimonious explanation across all three tracks is a supervision distribution/format mismatch in the generated-label pipeline, not a capacity ceiling in the student.

7 Limitations

Confounded comparison in §6. The gold-vs-generated contrast varies provenance (human-curated vs. LLM-generated), source-problem distribution (GSM8K/OmniMath vs. MATH-style traces), and label format simultaneously. This paper should not be read as isolating provenance. The missing controlled cell is generated teacher labels on the *same* GSM8K/OmniMath source problems used by the gold rows—the highest-value next experiment.

Seed and scale asymmetry. The generated-label baselines use one seed each from a ∼10-configuration sweep, while gold sources use four; more generated-label seeds could narrow the gap, though the breadth of the sweep makes a pure unlucky-seed explanation unlikely. The 1.5B (§5) and 3B (§6) tracks are not a controlled capacity ablation of one another—they differ in loss recipe and evaluation N—so "capacity doesn't fix it" is suggestive across tracks, not a single-variable ablation.

Not a SOTA PRM claim. The public Qwen2.5-Math-7B-PRM800K baseline is stronger than any single gold-source seed on the same split. The contribution is diagnostic: a format-matched cheap check can reveal supervision mismatch before investing in a larger student or a broader recipe search.

OlympiadBench, the positive control, and long-context. OlympiadBench is high-variance over two seeds (boundary diagnostic, not a headline source). The positive control uses a weak-teacher student whose score head is miscalibrated, so part of the measured gap may reflect degeneracy. And, as noted in §4.4, the collapse of the teacher effect on the hardest tier conflates mathematical intractability with long-context attention degradation; we cannot fully separate them here.

Downstream validation of §6 is incomplete. Unlike §5, the 3B gold-vs-generated result is evaluated only on step-level metrics; we do not show the gold-supervised 3B verifier beats majority vote on downstream best-of-N.

8 Conclusion

Under a fixed three-tier teacher probe, privileged (answer-aware) process supervision helps a teacher precisely in a tractability window—where it is both needed and usable—and this effect replicates across model families and within MATH by difficulty level. That advantage does not distill into a small student verifier: a 1.5B student trained on privileged labels is statistically indistinguishable from one trained on no-GT labels, and neither clears majority vote on downstream best-of-N. This null is not explained by an insensitive training pipeline, which we confirm with a positive control. Diagnosing further, we show that under an identical training/evaluation path, format-matched gold process supervision produces a competent verifier while our generated teacher labels do not, at both 1.5B and 3B scale—pointing at supervision distribution and format, not student capacity, as the dominant explanation. For efficient PRM development this suggests a practical ordering: before scaling students or re-running the same distillation loop, test whether the supervision distribution itself can support the target verifier.

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A Per-seed bootstrap gaps (§6.1)

Representative sequence-cluster bootstrap ROC-AUC gaps (2,000 resamples); the full 16-row per-seed table is released with the supplementary material.

Model A	Model B	ROC-AUC gap	95% CI (p)
GSM8K gold seed 0	best generated-label baseline	+0.111	[0.087, 0.134] (0.001)
OmniMath gold seed 0	best generated-label baseline	+0.121	[0.096, 0.146] (0.001)
GSM8K gold seed 0	generated privileged BCE	+0.193	[0.164, 0.221] (0.001)
OmniMath gold seed 0	generated privileged BCE	+0.203	[0.176, 0.232] (0.001)
Qwen PRM800K	OmniMath gold seed 3 (best seed)	+0.051	[0.033, 0.068] (0.001)

B Generated-label recipe ablations (§6)

Preliminary generated-label ablations on the Qwen2.5-3B score-head path.

Generated-label recipe	Priv ROC-AUC	No-GT ROC-AUC	Note
Score+critique MSE, early run	0.4582	0.4870	collapsed near all-negative
Score-only MSE	0.4844	0.4776	stable but weak
Feedback-token truncation fixed	0.4514	0.5093	stable but negative for priv.
BCE, error weight 3	0.5411	0.4700	best short privileged ablation, still weak
Rank-only, balanced batches	0.4939	0.6306	no-GT wins strongly
BCE+rank, balanced batches	0.5214	0.6140	no-GT wins strongly
BCE, error weight 3, longer training	0.4782	0.4971	longer BCE degrades signal

C Reproducibility note

All numbers trace to committed result artifacts released (anonymized) with the supplementary material. The teacher-level probe uses ProcessBench-style first-error F1; step-level student metrics use threshold-free ROC-AUC to avoid a fixed-cutoff calibration artifact. Paired-bootstrap CIs on step-level gaps resample paired step scores; Phase B gaps resample whole solutions (sequence-cluster bootstrap) because steps within a solution are correlated. The positive-control gap is recomputed directly from the committed per-step scores of the two conditions on the identical 6,505-step split. One codebase caveat we flag for reproducers: earlier ablation checkpoints and result files were written to shared, non-namespaced paths and overwritten across retrains; the numbers reported here are taken from the currently committed artifacts and recomputed where a historical log entry could not be matched to a specific committed file.